A small theory of Tweety

Vocabulary, assumptions, and an observation

1. Vocabulary

Thing: the domain of individuals.

Bird, Penguin, Flies: predicates with signature Thing -> Bool.

tweety: an individual of sort Thing.

The vocabulary introduces symbols; it does not assert their extension.

2. Starting points

A1 / AXIOM 1

All penguins are birds.

Assume a universal inclusion for this small theory.

A2 / AXIOM 2

No penguins are fliers.

Assume a strict exclusion, without exceptions, in this theory.

O1 / OBSERVATION

Tweety is a penguin.

Take this as an input fact; no empirical validation accompanies it.

Reading note

Axioms supply assumptions. An observation supplies another premise.

The next page states two consequences and two unresolved passages.

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Consequences and unresolved language

3. Stated consequences

L1 / LEMMA 1

Tweety is a bird.

Proof sketch: instantiate A1 at tweety; apply the observation O1.

T1 / THEOREM 1

Tweety is not a flier.

Proof sketch: instantiate A2 at tweety; apply the observation O1.

4. Deliberately unresolved passages

U1 / AMBIGUOUS GENERALIZATION

Birds are normally fliers.

Reading question: strict universal, statistical tendency, or default?

U2 / AMBIGUOUS REFERENCE

It is a flier.

Reading question: which individual does "it" refer to?

What a check would establish

A proof check concerns the exact formal conclusion and named premises.

A coherence check concerns compatibility of the selected formal claims.

Neither check validates the observations or resolves ordinary language.